

Introductory Calculus

ANSWER KEY



Direct substitution

- 1 Find $\lim_{x \rightarrow 2} (3x^2 - 2x + 1)$.
 $= 3(4) - 2(2) + 1 = 12 - 4 + 1 = 9$.
- 2 Find $\lim_{x \rightarrow -1} (x^3 - 5x)$.
 $= (-1)^3 - 5(-1) = -1 + 5 = 4$.

Indeterminate forms: factor and cancel

- 3 Find $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$.
Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 25}{x - 5} = \frac{(x - 5)(x + 5)}{x - 5} = x + 5$ (for $x \neq 5$), so the limit is $5 + 5 = 10$.
- 4 Find $\lim_{x \rightarrow -3} \frac{x^2 - 9}{x + 3}$.
Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 9}{x + 3} = \frac{(x + 3)(x - 3)}{x + 3} = x - 3$ (for $x \neq -3$), so the limit is $-3 - 3 = -6$.

The derivative as a limit

- 5 Write the difference-quotient definition of $f'(x)$.
$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$
- 6 Using Theorem 5.1 ($f(x) = x^2 \Rightarrow f'(x) = 2x$), find the slope of the tangent line to $f(x) = x^2$ at $x = 3$.
 $f'(3) = 2(3) = 6$.
- 7 Using Theorem 5.1, find the slope of the tangent line to $f(x) = x^2$ at $x = -5$.
 $f'(-5) = 2(-5) = -10$.

Differential equations

- 8 Show that $y = x^2 - 7$ solves $\frac{dy}{dx} = 2x$.
By Theorem 5.1, the derivative of x^2 is $2x$, and the derivative of the constant -7 is 0 . So $\frac{dy}{dx} = 2x + 0 = 2x$, matching the equation exactly.

- 9 Explain why solving a differential equation gives a family of functions rather than a single number.

A differential equation only constrains the derivative, not the exact function; any constant added to a valid solution y has derivative 0 and doesn't change whether it solves the equation, so every $y =$ (that solution) $+C$ also works — a whole family, not one answer.